Complex phase retrieval in dimension four needs exactly eleven measurements

Literature answer to the open minimum in arXiv:1605.08034

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Status: literature-already-answered, exact value. No ten measurement vectors can perform standard phase retrieval in $\mathbb{C}^4$, while an explicit eleven-vector frame does. Therefore

$$N_{\min}(\mathbb{C}^4) = 11.$$

Source problem

For vectors $a_1, \ldots, a_N \in \mathbb{C}^4$, standard phase retrieval asks that

$$x \longmapsto (|a_1^* x|^2, \ldots, |a_N^* x|^2)$$

be injective modulo multiplication of x by a scalar in the unit circle. Wang–Xu state on source PDF page 5 that the smallest possible N was unknown even for $d = 4$ [1]. Vinzant had already constructed an injective eleven-vector frame, whereas the best lower bound allowed ten.

Explicit later solution

Huang's Theorem 1.2 states that no family of ten vectors in $\mathbb{C}^4$ possesses the phase-retrieval property [2]. Corollary 1.3 combines this lower bound with Vinzant's explicit eleven-vector frame [3] and concludes

$$N_{\min}(\mathbb{C}^4) = 11.$$

The theorem and corollary appear on supporting PDF page 4; the lower-bound proof is on pages 10–14. Vinzant's frame and its injectivity certificate appear on the second supporting PDF, pages 1–2.

Proof intuition

Suppose ten vectors did phase retrieval. After Parseval normalization, their ten intensities sum to one, so they define an injective smooth map from $\mathbb{CP}^3$ into a nine-dimensional affine hyperplane. The rank-two kernel criterion for phase retrieval makes its differential injective; compactness then turns the map into a smooth embedding

$$G : \mathbb{CP}^3 \hookrightarrow \mathbb{R}^9.$$

Its oriented normal bundle ν has real rank three. The stable tangent–normal identity

$$T\mathbb{CP}^3 \oplus \nu \cong \underline{\mathbb{R}}^9$$

and the Euler sequence give $p_1(T\mathbb{CP}^3) = 4h^2$, hence $p_1(\nu) = -4h^2$.

Now use the special rank-one form of an intensity coordinate. For any nonzero measurement vector a_j, its zero locus is a projective hyperplane $H_j \cong \mathbb{CP}^2$. Along H_j, the corresponding centered coordinate direction is everywhere orthogonal to the embedded tangent space, yielding a nowhere-zero section of $\nu|_{H_j}$. Thus

$$\nu|_{H_j} \cong \underline{\mathbb{R}} \oplus \eta$$

for an oriented rank-two bundle η. Writing $\bar{h} = h|_{H_j}$, one side forces

$$p_1(\nu|_{H_j}) = -4\bar{h}^2,$$

while the splitting forces $p_1(\nu|_{H_j}) = p_1(\eta) = e(\eta)^2 = k^2\bar{h}^2$ for some integer k. The impossible equation $k^2 = -4$ rules out ten measurements. Vinzant's certified frame supplies eleven, closing the gap.

Scope boundary

This is an explicit later resolution, not an agent-generated theorem. It settles the source's standard rank-one question only in complex dimension four. It does not determine the separate minima for generalized Hermitian measurements, fusion-frame projections, arbitrary low-rank recovery, or complex dimensions $d \geq 6$ not covered by the known $d = 2^k + 1$ family.

References

[1] Y. Wang and Z. Xu, *Generalized phase retrieval: measurement number, matrix recovery and beyond*, arXiv:1605.08034 (2016/2019), p. 5.

[2] M. Huang, *Phase Retrieval in $\mathbb{C}^4$ Requires Exactly Eleven Measurements*, arXiv:2607.27719v1 (2026), Theorem 1.2 and Corollary 1.3.

[3] C. Vinzant, *A small frame and a certificate of its injectivity*, arXiv:1502.04656 (2015), Theorem 1.